

Gamified VR Educational Platform for Middle School Curriculum in Sri Lanka

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Our Team



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PP2 Comments & Actions Taken

Comment	Action Taken
UI near Sorowwa is not visible	Adjusted UI position and improved visibility
UI elements are overlapping	Fixed layout spacing and alignment
Animation issues	Optimized animation timing and transitions
Trees and models size incorrect	Standardized model scaling across environment
Questions not aligned with activity	Updated questions to match learning tasks



Problem in Brief

Students learn theory without practical experience

Limited access to labs and physical resources

Low engagement in traditional classroom learning

Difficulty understanding real-world STEM applications

Design Excellence of the Overall Solution

Feature	Existing Systems	Proposed System
Sri Lankan syllabus alignment	✗	✓
Hands-on VR task simulation	✗	✓
Gamified learning progression	⚠	✓
Practical skill assessment	✗	✓
Offline school usage	✗	✓

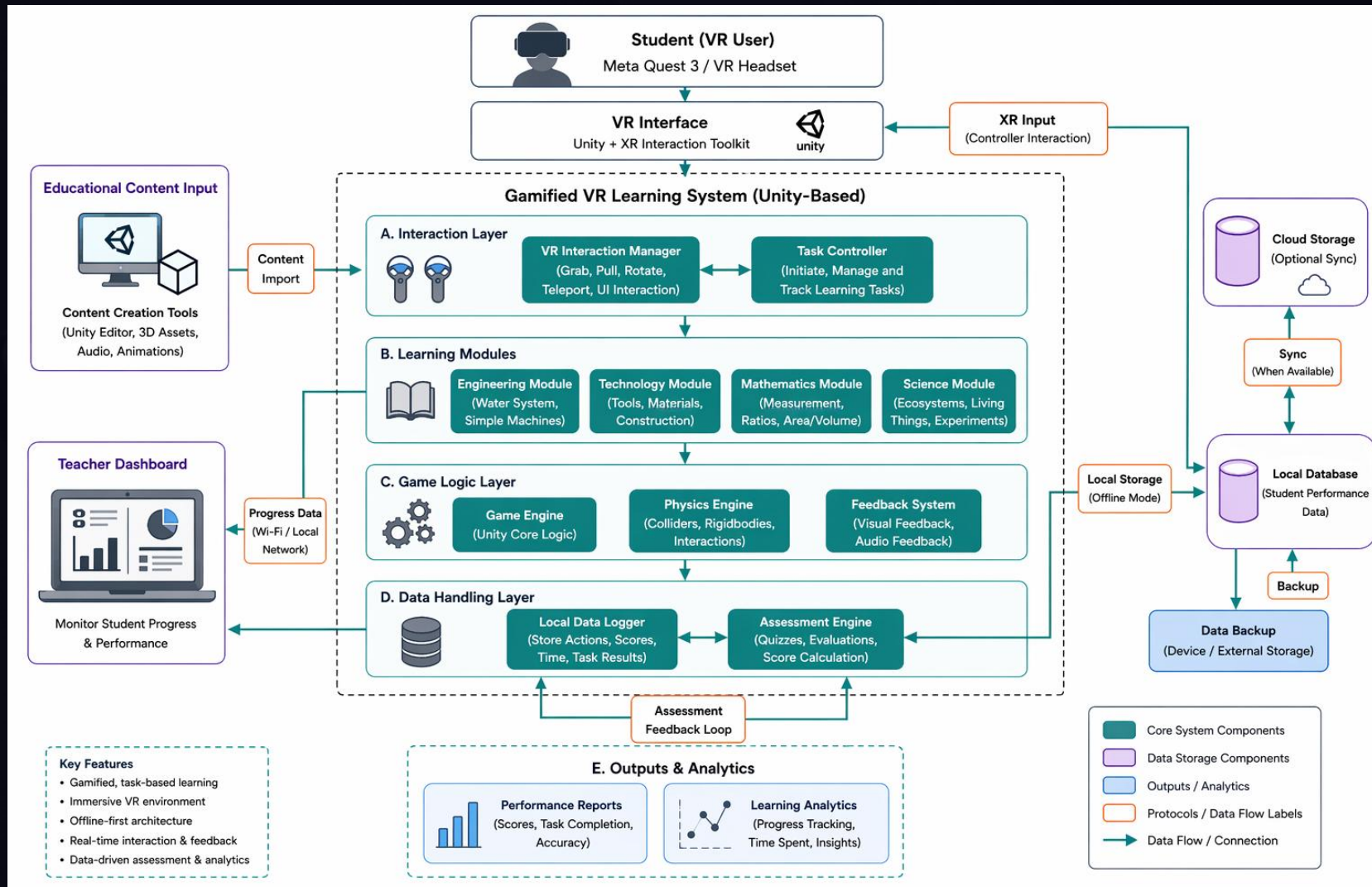
Most VR education tools are not aligned with the Sri Lankan curriculum

Many systems provide passive experiences (videos or demonstrations) instead of interactive learning

Existing platforms lack cultural and historical context

Limited gamification and immersive interaction

System Architecture & Integration



Mathematics Component



Description



Focuses on applying mathematical concepts through practical problem-solving activities in VR.

What Was Developed



- Sluice gate fraction control system
- Water level adjustment using fractions
- Balance scale for weight and equations
- Mathematics quiz and feedback system

Science Component



Description



Focuses on teaching scientific concepts through observation and experimentation in a virtual environment.

What Was Developed



- Food web ecosystem simulation
- Animal interaction and identification system
- Botanical inspection using magical lens
- Science quiz and feedback system

Technology Component



Description



Focuses on implementing interactive technology scenarios that simulate traditional Sri Lankan systems in VR.

What Was Developed



- Sekkuwa oil press interaction
- Atuwa pillar placement and puzzle solving
- VR object interaction using XR Toolkit
- Technology quiz and feedback system

Engineering Component



Description



Focuses on applying engineering principles through interactive mechanical systems in VR.

What Was Developed



- Lever-based lifting mechanism
- Mechanical advantage with construction simulation
- Force and load interaction system
- Engineering quiz and feedback system

Results & Evaluation

Tested with 20 students

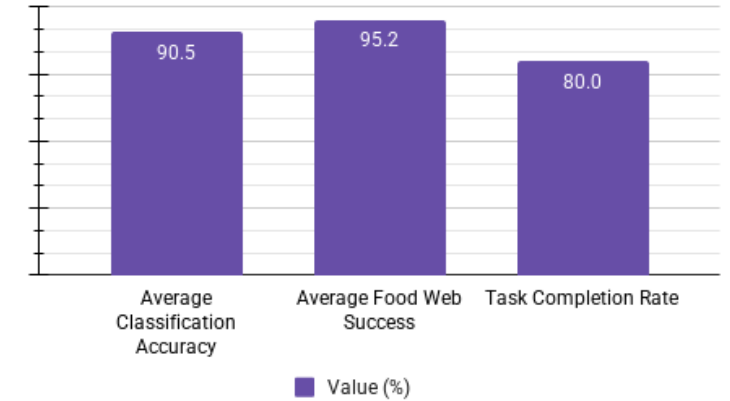
Increased task completion rate

Higher student engagement observed

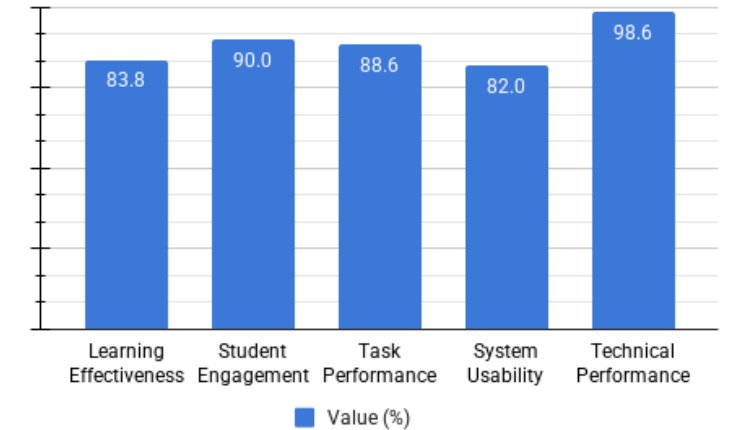
Improved understanding of STEM concepts

Positive feedback from teachers and students

Student Performance Metrics



Overall Evaluation Summary



Showcase of System Testing





Testimonials

“Learning feels interactive and engaging” – Student

“Helps students understand concepts faster” – Teacher

“More effective than traditional teaching methods” – Observation



Limitations and Future Improvements

Limitations

- Limited number of learning modules
- Requires VR hardware availability
- Small sample size for testing

Future Improvements

- Add more subjects and modules
- Expand virtual environments
- Enable multiplayer learning
- Integrate cloud-based data tracking





Thank You!

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